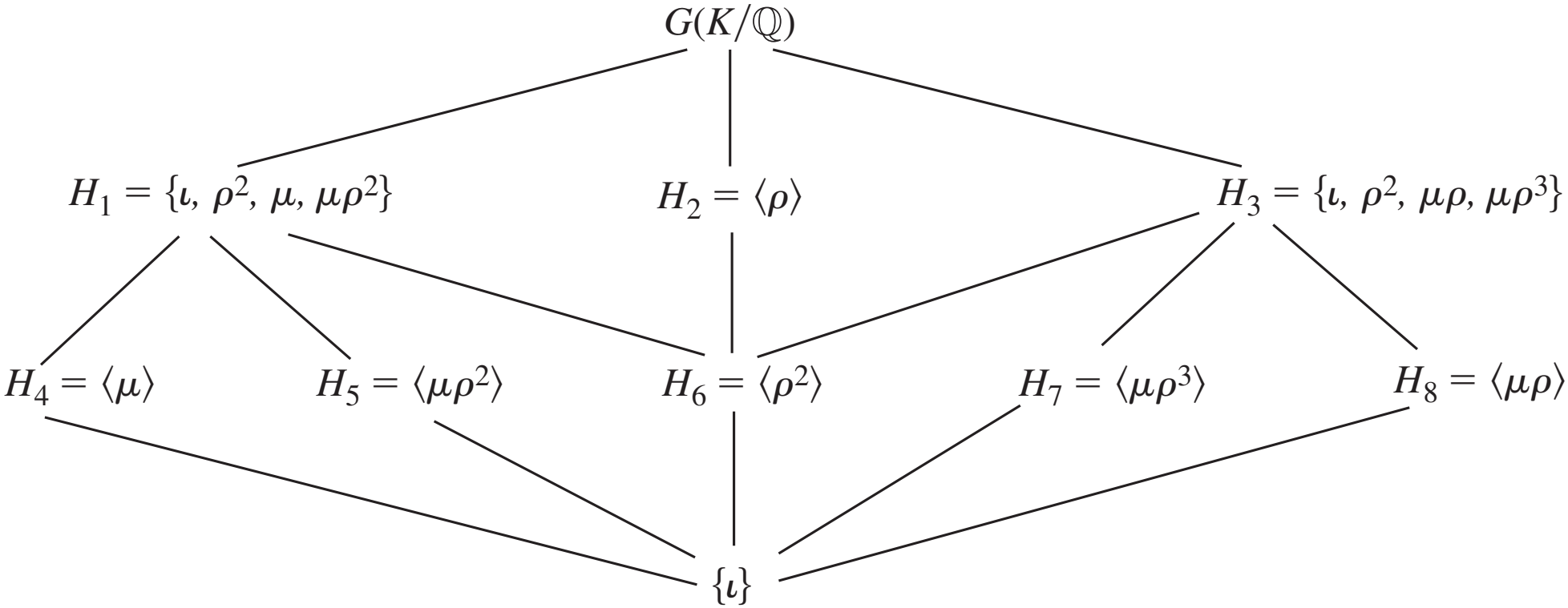

$G(K/\mathbb{Q})$
$H_1 = \{\iota, \rho^2, \mu, \mu\rho^2\}$
$H_2 = \langle\rho\rangle$
$H_3 = \{\iota, \rho^2, \mu\rho, \mu\rho^3\}$
$H_4 = \langle\mu\rangle$
$H_5 = \langle\mu\rho^2\rangle$
$H_6 = \langle\rho^2\rangle$
$H_7 = \langle\mu\rho^3\rangle$
$H_8 = \langle\mu\rho\rangle$
$\{\iota\}$

(a)